2.1 What "Eventually" Means

WORKSHEET

The question

If $s_1, s_2, s_3, \ldots$ and $t_1, t_2, t_3, \ldots$ are sequences of positive numbers, we write

$$(s_n) \leq (t_n)$$

to mean

"there exists a positive integer m such that $s_n \leq t_n$ whenever $n \geq m$".

Determine whether each of the following statements is true or false. In the case of a true statement, you should give a proof which includes an explicit determination of an appropriate m; in the case of a false statement, you should give a counterexample.

(i) $(1000n) \leq (n^2)$.

(ii) If it is not the case that $(s_n) \leq (t_n)$, then it is the case that $(t_n) \leq (s_n)$.

(iii) If $(s_n) \leq (t_n)$ and $(t_n) \leq (u_n)$, then $(s_n) \leq (u_n)$.

(iv) $(n^2) \leq (2^n)$.

Hints

Try the question on your own first. If you get stuck, read the hints *one at a time*, and go back to the question after each one. Hints marked *Pause and try* are the key steps.

Ask yourself The definition contains the phrase "there exists a positive integer m". To prove that $(s_n) \leq (t_n)$, what exactly do you have to produce? To prove that it is *false*, what do you have to show?

STAGE 2 Is "eventually smaller" like ordinary "smaller"?

Ask yourself What does "it is *not* the case that $(s_n) \leq (t_n)$" actually mean? Write it out in full, without the word "not" in front.

Pause and try Find two sequences of positive numbers that swap places for ever, so that neither is eventually smaller than the other.

STAGE 3 Chaining

Ask yourself In part (iii) you are given two values of m, one for each hypothesis. They might be different. Which one should you use for the conclusion?

STAGE 4 A quadratic against an exponential

Ask yourself To prove $n^2 \leq 2^n$ for all $n \geq 4$ by induction, you need to get from $n^2 \leq 2^n$ to $(n+1)^2 \leq 2^{n+1} = 2 \cdot 2^n$. What inequality between $(n+1)^2$ and n^2 would make that step work?

Try this next

1 **The specimen warm-up.** (From the STEP III Specimen paper, issued before the first STEP in 1987.) Show by using the binomial expansion that $(1+x)^n > nx$ whenever $x \geq 0$ and n is a positive integer. Deduce that if $y > 1$ then, given any number K, an integer N can be found such that $y^n > K$ for all $n \geq N$. Show further that if $y > 1$ then, given any K, an integer N can be found such that $y^n/n > K$ for all $n \geq N$. (*Hint for the last part:* use the next term of the binomial expansion.)

2 **Both ways round.** Prove that if $(s_n) \leq (t_n)$ and $(t_n) \leq (s_n)$, then $s_n = t_n$ for all sufficiently large n.

3 **Racing to infinity.** Find two *increasing* sequences of positive numbers, both tending

to infinity, neither of which is eventually smaller than the other.

4 Up the ladder. Show that $(2^n) \leq (n!)$, giving an explicit m. Then show that $(n^{10}) \leq (2^n)$ and find the smallest m that works. (A spreadsheet helps for the second part; the answer is surprisingly large.)

Further reading

- **Lara Alcock, *How to Study for a Mathematics Degree*** (Oxford University Press). Chapter 3 is about exactly this kind of logical language and why new undergraduates find it hard.
- **Lara Alcock, *How to Think About Analysis*** (Oxford University Press). A friendly guide to the definitions of first-year analysis.
- **Daniel Velleman, *How to Prove It*** (Cambridge University Press). A systematic introduction to logic, quantifiers and proof.

The full guided solution, and what the question leads to at university, is in Section 2.1:
https://keithmcnulty.github.io/step_book_mcnulty/ch02/eventually.html